

Graph Attention Transformer in IVF Outcome Prediction:

A Patient-Similarity Approach within the IVF Digital Twin Pipeline

Why Graphs? The Clinical Intuition Behind the Method

Every reproductive endocrinologist carries, consciously or not, a mental library of cases. When evaluating a new patient, the thought process is rarely purely algorithmic — it involves pattern recognition: *"I've seen patients like this before. Let me think about how those cycles turned out."* This is, in essence, what a graph-based model does — but systematically, at scale, and without the limitations of human memory.

The Graph Attention Transformer (GAT) layer within IVF Digital Twin formalises this intuition into a mathematically rigorous framework. Rather than treating each patient as an isolated data point, it constructs a network of clinical similarity across an entire cohort, allowing the model to propagate information between patients who share comparable biological and laboratory profiles. The result is a prediction that is not only informed by the individual's own parameters, but also contextualised by the observed outcomes of genuinely similar patients.

The Structure of a Patient Graph

At the heart of the method lies a **clinical similarity graph**. Each patient in the training cohort is represented as a node — a point in an 18-dimensional space defined by clinical and embryological features: patient age, antral follicle count, number of oocytes retrieved, fertilisation outcomes, blastocyst counts, blastocyst quality rates, embryo cryopreservation figures, and laboratory efficiency indices including the KPI Score.

Crucially, the graph topology — that is, which patients are connected to which — is determined exclusively by these clinical parameters. The KAT neural network score (PRAI) is deliberately excluded from the similarity calculation. This design choice is intentional: the graph must reflect biological and laboratory resemblance, not the prior beliefs of another model. Edges between patients are weighted by cosine similarity, with stronger connections drawn between patients whose full clinical profiles align most closely.

The result is a structured representation of the cohort in which proximity carries real clinical meaning: two nodes connected by a thick edge represent two patients who arrived at their IVF cycles with similar ovarian reserve, responded similarly to stimulation, and produced comparable embryological outcomes in the laboratory — regardless of whether their final transfer results were the same.

How the Model Learns

The Graph Attention Transformer architecture processes this network through a mechanism of **message passing with learned attention**. In each computational layer, every node aggregates information from its neighbours, weighting their contributions according to an attention score that the model learns during training. Neighbours who share stronger clinical resemblance and whose features are more informative for the prediction task receive higher attention weights.

This is distinct from conventional neural networks, which process each patient independently. Here, the representation of a patient is iteratively refined by what is known about similar patients — their features, their relationships within the cohort, and ultimately their outcomes. After three such layers of message passing, the model produces an embedding for each patient that encodes both individual characteristics and neighbourhood context.

Two task-specific output heads then translate these embeddings into predictions: one for the probability of clinical pregnancy following embryo transfer, and one for the continuous PRAI score. This multi-task formulation means the model is simultaneously learning to classify outcomes and to align its internal representations with the continuous probability estimates generated by the KAT ensemble — reinforcing coherence across the pipeline layers.

Training on Real Clinical Protocols

The model was trained on **1,172 real clinical IVF protocols** collected from a single centre. All cycles included complete embryological cascade data from ovarian stimulation through to blastocyst culture, with documented transfer outcomes. The dataset reflects the natural diversity of an IVF clinical population: patients ranged in age, ovarian reserve, stimulation response, and fertilisation efficiency, with a pregnancy rate of 54% across the cohort.

Training followed a **transductive cross-validation design**: a single graph was constructed on the full cohort, and in each of five folds, the outcome labels of the validation patients were masked before training began — but those patients remained visible within the graph structure. This mirrors the intended use of the model in practice, where a new patient joins the existing network of known cases, benefits from the information propagated through connected neighbours, yet contributes no outcome label to the training signal. The model is never evaluated on data it was trained on.

Key training decisions:

- Graph topology built on 18 clinical features; KAT scores included only in node embeddings, not in edge construction — preventing the graph structure from reflecting existing model predictions rather than clinical biology.
- Label smoothing ($\epsilon = 0.08$) was applied to regularise the binary outcome targets, preventing the model from becoming overconfident in ambiguous cases.
- L2 weight regularisation and attention dropout were tuned specifically to reduce the gap between training and cross-validation performance.

- The cosine warm-up learning rate schedule allowed stable initialisation of attention weights before the full learning rate was applied.

Validation Metrics

Cross-validation performance was assessed using out-of-fold predictions — each patient's prediction was generated by a model that had never seen that patient's outcome label during training.

Metric	Graph Transformer (GNN)	KAT Ensemble (PRAI)	GAT+KAT Ensemble
AUC-ROC	0.632	0.652	~0.658–0.665*
Brier Score	0.241	0.233	~0.229*
F1 Score	0.623	—	—
Optimal threshold	0.49	—	—

**Ensemble metrics depend on the cohort-specific optimal weight w_{GNN} , determined automatically during cross-validation.*

The Graph Transformer alone does not surpass the KAT ensemble in discriminative power. This is expected: the KAT model was built and calibrated on a substantially larger dataset, and benefits from more mature feature engineering. What the graph model contributes is **complementary signal** — a different computational perspective on the same patient, derived from neighbourhood structure rather than individual feature weighting.

The correlation between GNN and KAT predictions across the cohort is approximately $r = 0.79$. This means the two models agree substantially but not completely — they disagree in a structured, non-random way that reflects their methodological differences. It is precisely this partial disagreement that makes their ensemble valuable: when two models diverge on a case, combining their outputs reduces variance and improves calibration.

The Ensemble: Why Combining Both Models Matters

The final GAT Ансамбль score is a weighted combination of the Graph Transformer and the KAT neural network prediction:

$$P(\text{pregnancy}) = w_{\text{GNN}} \times P_{\text{GNN}} + (1 - w_{\text{GNN}}) \times P_{\text{KAT}}$$

The weight w_{GNN} is not set manually — it is determined during cross-validation by identifying the combination that maximises AUC on out-of-fold predictions. In practice, the optimal weight typically falls between 0.25 and 0.40, reflecting the relative contributions of each model to the ensemble.

This design philosophy follows a well-established principle in machine learning: **diverse models make better ensembles**. The KAT network processes each patient in isolation, building a prediction from individual feature patterns learned across thousands of examples. The Graph Transformer, by contrast, anchors its prediction in the observed landscape of similar patients — it asks not just *"what do this patient's numbers suggest?"* but *"how do patients with profiles like this one typically fare?"*

The ensemble gains from both: the KAT model's broad statistical learning and the graph model's ability to leverage local cohort structure. In cases where the KAT model is uncertain — for example, patients whose clinical profiles lie at the boundary between response categories — the graph signal from genuinely similar neighbours can provide meaningful supplementary evidence.

The Unique Clinical Value: Real Patients, Real Context

Perhaps the most clinically distinctive feature of the GAT layer is its interpretability through patient similarity. When the model produces a prediction, it can identify — and display — the ten patients from the training cohort whose clinical profiles most closely resemble the current patient, ranked by cosine similarity.

These are not synthetic cases, statistical averages, or model constructs. They are real patients, from real IVF cycles, with real documented outcomes. The clinician can observe not only the predicted probability but also the distribution of GNN-estimated probabilities among these neighbours — a direct, intuitive answer to the question: *"Among the patients most similar to this one, what did the model predict for them, and how variable was that prediction?"*

A patient whose nearest neighbours cluster tightly around high predicted probabilities provides a different kind of clinical confidence than one whose neighbours span the full range of outcomes. This contextualisation is something that no scalar probability score — however well-calibrated — can offer on its own.

Practical Considerations and Current Limitations

The model operates entirely on CPU and requires no specialised hardware. Inference for a single new patient, including graph construction and forward pass across the full training cohort, completes in under two seconds on a standard clinical workstation. The trained model is saved as a portable checkpoint file and integrates directly into the IVF Digital Twin application at startup.

The training cohort of 1,172 protocols is sufficient for a graph model of this scale — graph neural networks are known to be relatively data-efficient compared to large sequential models, because each patient's prediction is informed not only by its own features but by the entire neighbourhood structure. Nevertheless, the cross-validation AUC of 0.632 for the standalone graph model reflects an inherent ceiling imposed by the available signal in the input features. IVF outcome prediction is a genuinely hard problem; the irreducible biological variability of

embryo implantation means that no model — regardless of architecture — can be expected to achieve perfect discrimination.

The value of the GAT layer therefore lies not in replacing existing prediction approaches but in extending them: adding a clinically grounded, structurally distinct perspective that, when combined with the KAT ensemble, produces a more robust and better-calibrated final score.

Position Within the IVF Digital Twin Architecture

The GAT Graph Transformer constitutes Layer 6 (L6) of the IVF Digital Twin pipeline. It operates downstream of the KAT neural ensemble (L3) and receives the KAT probability score as one of its node features — not as a structural input to the graph, but as a node-level attribute that the attention mechanism can weight alongside clinical parameters. This positioning means the graph model has access to the KAT ensemble's learned summary of the patient while remaining free to form its own structural judgement about similarity and neighbourhood outcomes.

Together, the six layers of IVF Digital Twin provide a multi-perspective, multi-methodology assessment of IVF outcome probability — Monte Carlo stochastic simulation, ensemble pregnancy scoring, neural network prediction, cluster-based profiling, diffusion-based laboratory forecasting, and patient similarity graph analysis — each contributing an independent signal, and each capable of flagging cases where the consensus is strong or where uncertainty warrants closer clinical attention.

IVF Digital Twin v6.2 · Sergeev et al., 2025 · Research prototype — not for standalone clinical use